

Assignment 6: Triage Pseudocode

START

Step 1: Record intake data

Collect: chief complaint, onset/duration, past medical history, allergies, medication, arrival mode.

Measure: Temperature (Temp), heart rate (HR), respiratory rate (RR), blood pressure (BP), peripheral oxygen saturation (SpO2), blood glucose, weight, pain score

[Justification: These fields match those on the simplified version of the front sheet of the Mercer paper intake record]

Step 2: Screen for life-threatening condition (ESI 1)

IF(

Cardiac arrest OR

Major trauma OR

Shock OR

GCS $\leq$ 8

) THEN:

Assign ESI 1

Immediately send to resuscitation bay

Skip to Step 5

ELSE:

Continue to Step 3

[Justification: these conditions require immediate physician intervention. Waiting even 10 min risks irreversible harm.]

Step 3: Screen for high-risk / danger-zone vitals (ESI 2)

IF(

Chest pain OR

Suspected stroke OR
SBP < 90 mmHg OR SBP > 200 mmHg OR
MAP > 60 mmHg OR
HR < 40 bpm OR HR >140 bpm OR
SpO2 < 92% on room air OR
RR < 6 breaths/min OR RR > 30 breaths/min OR
Pain score >= 8 OR
GCS is 9 to 12

) THEN:

Assign ESI 2 (Emergency)
See physician within 10 mins

Skip to Step 5

ELSE:

Continue to Step 4

[Justification: SBP < 90 indicates shock. SBP > 200 indicates hypertensive emergency. MAP < 60 signals end-organ perfusion failure. SpO2 < 92% corresponds to approximately 60 mmHg of PaO2, the steep part of the oxygen-hemoglobin dissociation curve. These are standard critical-care thresholds.]

Step 4: Count required clinical resources (ESI 3 vs 4 vs 5)

IF (resources needed >= 2 (e.g. sepsis, abdominal pain)) THEN:

Assign ESI 3 (Urgent)
See physician within 30 mins

ELSE IF (resources needed = 1 (simple laceration, urinary tract infection)) THEN:

Assign ESI 4 (Semi-urgent)
See physician within 60 mins

ELSE:

Assign ESI 5 (Non-urgent)

Step 5: Document and zone

Record ESI level and re-triage time

STOP

Assumptions:

1. Strictly ordered (There is no backtracking once ESI is assigned)
2. Re-triage is a separate, continuous process. Any deterioration restarts the algorithm from Step 1.
3. Pediatric and obstetric patients use the same ESI logic in this ED (Mercer has no separate pediatric ED).

Clinical Context Write-Up

Setting: Mercer General is a 24/7 Caribbean regional ED serving approximately 80,000 residents plus seasonal tourists. It operates with roughly a dozen acute cubicles, two resus bays, a 4-bed short-stay bay, and a 6-chair fast-track area. These are all shared with the rest of the hospital's lab, radiology, and pharmacy. On heavy days the census exceeds 150 patients. There is no separate pediatric ED.

Why ESI fits this setting: The ESI 5-level system matches Mercer's resource reality well because the system classifies patients not just by acuity but also by resource consumption. This is seen in the distinction between ESI 3, 4, and 5. In a small ED with limited resources, knowing how many resources a patient will consume is as important as knowing how sick they are.

Key constraints the algorithm must respect:

Time pressure: Triage assessment takes 3–5 minutes. A nurse conducting the triage process would have no lab results, no imaging, and a limited patient history. Every threshold in the algorithm must be evaluable at the bedside with just basic vital signs monitored and a brief verbal assessment.

Friction budget: As Sister Patrice notes, a nurse at 03:00 with six waiting patients cannot fill an unbounded form. The algorithm is deliberately sequential with five binary questions, no branching back. This minimizes cognitive load and documentation time.

Re-triage reality: Mercer's waiting room holds approximately 20 seats. A patient triaged ESI 4 who deteriorates while waiting must be caught by the triage nurse's visual scan, not by the algorithm itself.

Exploratory Data Analysis (EDA)

This EDA section describes what *should* be captured and what analyses would matter most, aligned with the four things Sister Patrice wants us to understand

Variables to capture per encounter: Arrival timestamp, triage timestamp, seen-by-physician timestamp, disposition timestamp, exit timestamp; ESI level assigned; chief complaint (free text that is coded); age; sex; arrival mode (ambulance/walk-in/EMS); vital signs at triage; pain score; ESI level at re-triage (if changed); final diagnosis; disposition (admit/discharge/transfer); length of stay.

Key EDA questions to be explored:

1. What proportion of encounters fall into each ESI level?
2. Are patients consistently seen within the target time-to-be-seen suggested by the ESI level?
3. How often are patients escalated from their initial ESI level?
4. Are there any season or time-of-day patterns to the ESI levels of patients?
5. Do patients assigned to ESI 3 actually consume 2 or more resources?

AI Tool Engagement Log

Claude (Anthropic) was utilized in the creation of this submission. The questions asked to it were:

1. "Given the following vital signs, what vital sign thresholds should I use to distinguish ESI 1, ESI 2 and ESI 3?"
2. "Given the following feedback, suggest ways to improve this assignment?"